

■ PYTHON TEST PREPARATION NOTES

From Python Basics to Loops

Simple English • Quick Revision • Practice Questions • MCQs

Overview: These notes cover the Python topics learned so far: printing, variables, input, integers, calculations, percentage, average, if/elif/else, grades, complete marksheet process, for loops, range(), same-line output, and even/odd numbers. The final section is for test practice.

1. print()

Used to display text or a value on the screen.

```
print("Hello")
print(50)
```

2. Variables

A variable stores a value.

```
name = "Azhar"
age = 17
marks = 80
```

Text uses quotes. Numbers do not need quotes.

3. Basic Math Operators

Operator	Meaning	Example
+	Addition	10 + 5
-	Subtraction	10 - 5
*	Multiplication	10 * 5
/	Division	10 / 5
%	Remainder	10 % 3
**	Power	2 ** 3

4. input() and int()

input() takes a value from the user. int() converts suitable input into an integer.

```
name = input("Enter your name: ")
marks = int(input("Enter marks: "))
```

5. Percentage and Average

Percentage:

```
percentage = total / total_marks * 100
```

Average:

```
average = total / number_of_subjects
```

6. if / elif / else

if checks a condition. elif checks another condition. else runs when previous conditions are false.

```
if percentage >= 90:  
    print("A1")  
elif percentage >= 80:  
    print("A")  
else:  
    print("Fail")
```

7. Complete Marksheet Process

Step 1: Take student name.

Step 2: Take marks of each subject using `int(input())`.

Step 3: Add all subject marks to get total.

Step 4: Calculate percentage using `total / total_marks * 100`.

Step 5: Use `if / elif / else` to calculate the grade.

Step 6: Print the complete result: name, subject marks, total, percentage, and grade.

Example structure:

```
name = input("Enter name: ")  
english = int(input("English: "))  
physics = int(input("Physics: "))  
chemistry = int(input("Chemistry: "))  
math = int(input("Math: "))  
computer = int(input("Computer: "))  
  
total = english + physics + chemistry + math + computer  
percentage = total / 500 * 100  
  
# Then use if / elif / else for grade  
# Finally print the complete marksheet
```

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8. for Loop

A for loop repeats a task.

```
for i in range(1, 6):  
    print(i)
```

Output: 1, 2, 3, 4, 5

9. What is i?

i is a variable. During the loop, its value changes according to the range.

10. range()

range(1, 6) starts at 1 and stops before 6, so it gives 1 to 5.

11. Same-Line Output

```
for i in range(1, 10):  
    print(i, end="")
```

Output: 123456789

12. Even Numbers

```
for i in range(1, 11):  
    if i % 2 == 0:  
        print(i)
```

Output: 2, 4, 6, 8, 10

13. Odd Numbers

```
for i in range(1, 11):  
    if i % 2 != 0:  
        print(i)
```

Output: 1, 3, 5, 7, 9

Quick Revision Rules

- Use quotes for text.
- Use `int(input())` when you need a whole number from the user.
- Use `:` after if, elif, else, and for statements.
- Indent the code inside if/elif/else and loops.
- `range()` stops before its ending number.
- `%` gives the remainder.

■ TEST PRACTICE — QUESTIONS

Answer these yourself before checking any help.

1. What is the purpose of `print()` in Python?
2. What is a variable?
3. What is the difference between text and a number in Python?
4. What does `input()` do?
5. Why do we use `int(input())` for marks?
6. Write the Python operator for multiplication.
7. Write the formula for percentage.
8. Write the formula for average.
9. What is the purpose of `if`?
10. What is the purpose of `elif`?
11. What is the purpose of `else`?
12. What does a `for` loop do?
13. What is `i` in a `for` loop?
14. What does `range(1, 6)` produce?
15. What does `end=""` do in `print()`?
16. How can you print even numbers from 1 to 10?
17. How can you print odd numbers from 1 to 10?
18. Write the steps of a complete marksheet program in the correct order.
19. Write a program that takes five subject marks and calculates total and percentage.
20. Add a grade system using `if` / `elif` / `else` to your marksheet program.

■ MCQs — Quick Test

1. Which function displays output?

- A) `input()`
- B) `print()`
- C) `int()`
- D) `range()`

2. Which operator is used for multiplication?

- A) `+`
- B) `/`
- C) `*`
- D) `%`

3. Which function takes input from the user?

- A) `print()`
- B) `input()`
- C) `range()`
- D) `if()`

4. Which converts suitable text input to an integer?

- A) `str()`

- B) float()
- C) int()
- D) print()

5. What does 10 % 3 return?

- A) 3
- B) 1
- C) 0
- D) 10

6. Which statement is used to check a condition?

- A) if
- B) for
- C) print
- D) input

7. Which keyword checks another condition?

- A) else
- B) elif
- C) next
- D) check

8. What does range(1, 5) give?

- A) 1,2,3,4
- B) 1,2,3,4,5
- C) 0,1,2,3,4
- D) 2,3,4,5

9. Which one is an even number?

- A) 7
- B) 9
- C) 12
- D) 15

10. Which one is an odd number?

- A) 2
- B) 8
- C) 10
- D) 11

11. Which keyword is used for repetition over a range?

- A) for
- B) grade
- C) input
- D) total

12. What does end="" usually do?

- A) Stops Python
- B) Keeps output on the same line
- C) Adds a space only
- D) Deletes output

13. Which is a valid variable?

- A) my name
- B) 2marks
- C) marks
- D) class

14. Which formula correctly calculates percentage?

- A) $\text{total} * \text{total_marks}$
- B) $\text{total} / \text{total_marks} * 100$
- C) $\text{total} + 100$
- D) $\text{total_marks} / \text{total}$

15. Which order is correct for a marksheet?

- A) Grade → Input → Total
- B) Input → Total → Percentage → Grade → Output
- C) Output → Grade → Input
- D) Percentage → Input → Total

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Keep practicing. First understand the concept, then write the code yourself.

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